

EXPERIMENT NO: 10

AIM OF THE EXPERIMENT: - TO VERIFY THEVENIN'S THEOREM.

THEORY: -

Thevenin's theorem

In any linear active bilateral network the response in an element or a part of the network can be found out by using Thevenin's theorem.

Explanation: -

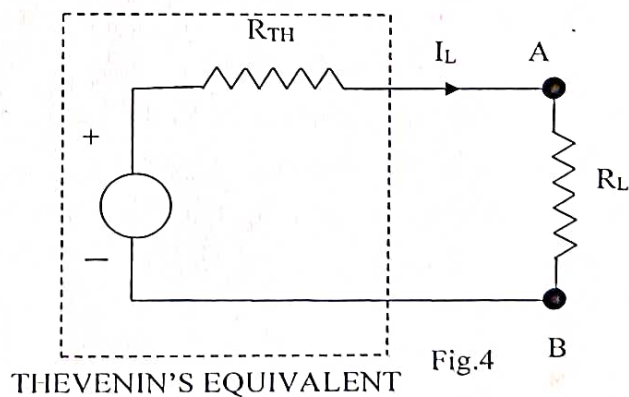
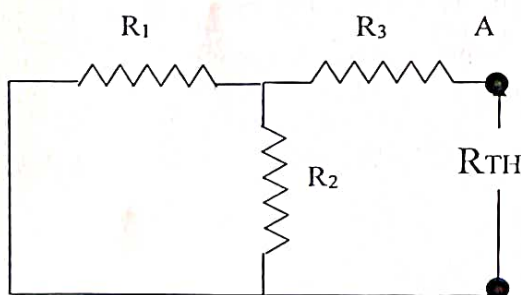
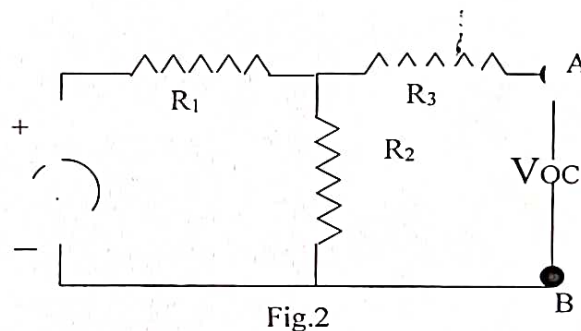
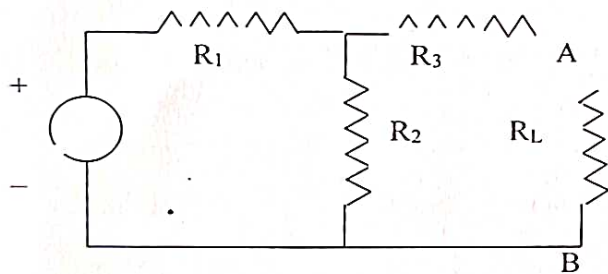
To explain the application Thevenin's theorem let us consider a simple circuit as shown in fig 1 to find the current in resistive load R_L as shown in fig 1.

Step-1 First of all remove the element from the circuit across which the response is required i.e R_L as shown in fig

APPARATUS REQUIRED

SL NO	APPARATUS	SPECIFICATION	QUANTITY
1	THEVENIN'S THEOREM KIT		01
2	PATCH CHORDS	4mm	AS REQUIRED

CIRCUIT DIAGRAM: -



CIRCUIT DIAGRAM FOR FINDING V_{oc} , I_L AND R_L :- (as per the kit)

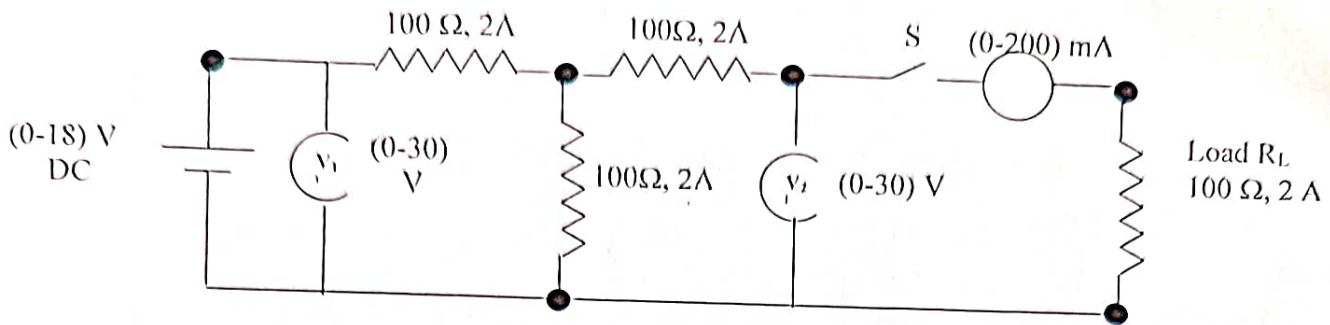


Fig-5

CIRCUIT DIAGRAM FOR FINDING R_{TH}

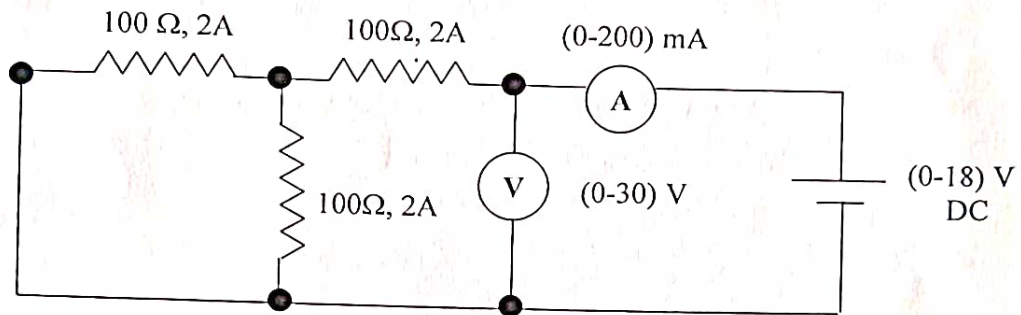


Fig - 6

Step-2 Find the open circuit voltage V_{oc} across the open terminals AB with the sources. The current through R_2 is given by:-

$$I_1 = V / (R_1 + R_2)$$

$$\therefore V_{oc} = I_1 \cdot R_2 = V R_2 / (R_1 + R_2)$$

The Thevenin's voltage V_{TH} is given by:

$$V_{TH} = V_{oc} = V R_2 / (R_1 + R_2)$$

Step-3 Replace the sources by internal resistances with source zero. Find the equivalent Thevenin's resistance R_{TH} across the terminals AB as shown in fig 3.

$$R_{TH} = R_3 + (R_1 \cdot R_2) / (R_1 + R_2) \text{ (since, } R_1 \text{ \& } R_2 \text{ are in parallel)}$$

Step-4 Draw the Thevenin's equivalent circuit as shown in fig 4 by connecting the load resistance ' R_L ' across the terminal A-B & find the response as shown in fig 4.

Therefore, the current ' I_L ' through ' R_L ' is given by:-

$$I_L = V_{TH} / (R_{TH} + R_L)$$

PROCEDURE:-

Procedure for finding V_{oc} , I_L & R_L :-

- Make connections as per the circuit diagram shown in fig 5.

- ii. Switch on the supply
 - iii. Open switch S & measure V_2 which is V_{oc}
 - iv. Close switch S & take the reading of voltmeter V_2 & ammeter reading I_L . $R_L = V_2/I_L$
- Procedure for finding R_{TH} :-**

- i. Make connections as per the circuit diagram in Fig 6.
- ii. Switch on the supply voltage.
- iii. Take the reading of voltmeter & ammeter.

TABULATION:-

For V_{oc} and R_L

Sl. No	Voltmeter meter reading (V_1) in volt	Ammeter reading (I_L) in Amp	Thevenin's Voltage $V_{oc}(V_2)$ in Vo	Load resistance $R_L = (V/I_L)$ In Ω
1				

For R_{TH}

Sl. No	Voltmeter Reading (V_2) In Volt	Ammeter Reading (I) In Amp	Thevenin's resistance $R_{TH} = (V/I)$ in Ω
1			

RESULTS & CONCLUSION :-

THEORITICAL VALUS: -

PRACTICAL VALUE: -

FROM THE ABOVE EXPERIMENT THE THEVENIN'S THEOREM WAS VERIFIED.

PRECAUTION: -

- i. ALL THE CONNECTIONS SHOULD BE RIGHT AND TIGHT.
- ii. THE AMMETER AND VOLTMETER SHOULD BE CAREFULLY CHOSEN SO THAT THEIR RANGES ARE MORE THAN MAXIMUM VALUES TO BE MEASURED.
- iii. SUPPLY SHOULD BE SWITCHED ON AFTER ENSURING CORRECTNESS OF CONNECTIONS.
- iv. ANY LIVE TERMINAL SHOULD NOT BE TOUCHED WHILE THE SUPPLY IS ON

QUESTIONS: -

1. State Thevenin's theorem?
2. How will you find out Thevenin's equivalent resistance?
3. What do you mean by open circuit voltage?
4. What is the utility of Thevenin's Theorem?
5. What is the condition for voltage source zero?
6. What is the condition for current source zero?